

```

1  #include <stdio.h>
2
3  #define MAX 20
4
5  int adj[MAX][MAX], indegree[MAX], queue[MAX];
6  int n;
7
8  void topologicalSort() {
9      int front = 0, rear = 0;
10
11
12      for (int i = 0; i < n; i++) {
13          indegree[i] = 0;
14          for (int j = 0; j < n; j++) {
15              if (adj[j][i] == 1)
16                  indegree[i]++;
17          }
18      }
19
20      for (int i = 0; i < n; i++) {
21          if (indegree[i] == 0)
22              queue[rear++] = i;
23      }
24
25      printf("Topological Order: ");
26
27      while (front < rear) {
28          int u = queue[front++];
29          printf("%d ", u);
30
31          for (int v = 0; v < n; v++) {
32              if (adj[u][v] == 1) {
33                  indegree[v]--;
34
35                  if (indegree[v] == 0)
36                      queue[rear++] = v;
37              }
38          }
39      }
40  }

```

```
41
42 int main() {
43     printf("Enter number of vertices: ");
44     scanf("%d", &n);
45
46     printf("Enter adjacency matrix:\n");
47     for (int i = 0; i < n; i++)
48         for (int j = 0; j < n; j++)
49             scanf("%d", &adj[i][j]);
50
51     topologicalSort();
52
53     return 0;
54 }
55
```

Enter number of vertices: 6

Enter adjacency matrix:

0 0 1 1 0 0

0 0 1 0 1 0

0 0 0 0 1 1

0 0 0 0 0 1

0 0 0 0 0 1

0 0 0 0 0 0

Topological Order: 0 1 3 2 4 5

Process returned 0 (0x0) execu

Press any key to continue.